# Same Score, Different Evidence: Decodability, Surface Sufficiency, and Causal Relevance in Code Models

**Anonymous Author(s)**
Affiliation
Address
email

## Abstract

Interpretability experiments must distinguish model mechanisms from measurement artifacts. We present a case-study-backed reporting matrix for linear probes over five identifier properties and roles in code models. The study includes three models and collectively covers all seven XLCoST programming languages without forming a complete factorial design. The matrix binds each claim to an estimand, comparator, matching unit, uncertainty statement, outcome, and falsifier. Two matched cases motivate it. On Python, boolean occurrence type is decodable at $0.981$–$0.988$ macro-F1, while a masked source-line classifier reaches $0.983$ without a language model. Paired problem-clustered deltas are $-0.003$ to $+0.002$, with intervals excluding probe advantages above $0.014$–$0.017$, although the sign of that difference varies by language and by comparator. A full-residual patch at a class or structure query-name site also fails its specified bidirectional gate in Qwen2.5-1.5B, recovering $0.009$ to $0.020$ of the matched behavioral gap. This bounds the effect at that site rather than showing global non-use. A third case exposes a failed control. Role-conditioned iterator renaming yields $0.998$–$0.999$ F1 at embedding layer zero, consistent with a lexical code introduced by the intervention rather than semantic robustness. Decodability, capacity control, cue sensitivity, surface sufficiency, and site-state causal relevance therefore require separate, outcome-aware records. We make the implementation of our methods accessible at `anonymous.4open.science/r/SameScoreDifferentEvidence2026`.

## 1 Introduction

Linear probes are often used to ask what information a representation contains. A successful probe establishes decodability under a particular readout. It does not by itself establish that the model computed an abstract feature, that the feature is unavailable in surface form, or that the model uses it behaviorally [Belinkov, 2022, Voita and Titov, 2020]. Control tasks constrain probe capacity [Hewitt and Liang, 2019], and targeted studies show that probe accuracy need not entail task relevance [Ravichander et al., 2021, Elazar et al., 2021]. Capacity is only one source of ambiguity. Input leakage, label construction, sample resolution, and an unmatched intervention can each preserve high accuracy while changing the scientific claim.

This distinction is central to interpretability as a science. A measurement is useful when its target is explicit, alternative explanations are tested, and contrary outcomes change the conclusion. We use identifier probing as a comparative case study. Some targets are semantic variable roles, such as accumulating values or indexing a container [Sajaniemi, 2002]. Others are occurrence types or declared type identifiers. Names, syntax, and context can all predict these targets, which makes probe accuracy underidentified. Prior code-probing studies, including the 15-task INSPECT framework,

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Table 1: Target coverage. No row should be read as a complete three-model by seven-language experiment. Table 2 gives the outcome-aware evidence records.

| Target | Models | Language coverage |
|---|---|---|
| Accumulator | Qwen2.5-1.5B | Python plus Java, C++, C, C# |
| Index or key | Qwen2.5-1.5B | Python, Java, JavaScript, C++, C, C# |
| Iterator | All three | Python, Java, JavaScript, PHP, C++, C, C# |
| Boolean | All three | Python, Java, JavaScript, PHP |
| Class or structure | All three | Python, C++, JavaScript, C |

map decodable syntactic and semantic information across layers and models [Karmakar and Robbes, 2021, Troshin and Chirkova, 2022, Karmakar and Robbes, 2024]. We do not propose another probe, control, or task taxonomy. Existing work catalogs probing limitations and shows that activation-patching conclusions depend on metric and intervention choices [Belinkov, 2022, Zhang and Nanda, 2023, Heimersheim and Nanda, 2024]. Subspace interventions can also create an illusion of localization [Makelov et al., 2023], which motivates calls for falsifiable interpretability designs [Leavitt and Morcos, 2020] and for naming the causal mediator a claim is about [Mueller et al., 2024]. Reporting standards in other empirical sciences answer a related need by binding a published claim to its estimand, comparator, and uncertainty statement [Schulz et al., 2010]. Our distinct contribution is an outcome-aware reporting unit that binds each claim to its estimand, comparator, matching unit, uncertainty, observed outcome, and falsifying next test. Unlike a control checklist, the record preserves failed and confounded comparisons and states how they change the licensed claim.

Our evidence covers five roles, three models, and the seven-language XLCoST corpus [Zhu et al., 2022]. The experiment is intentionally reported as an uneven evidence matrix rather than a full factorial study. Qwen2.5-1.5B, Qwen2.5-Coder-1.5B, and StarCoder2-7B all appear in the controlled boolean, class or structure, and iterator experiments [Qwen et al., 2024, Hui et al., 2024, Lozhkov et al., 2024]. Earlier accumulator and index runs use one model and a weaker split protocol. Iterator runs cover all three models and all seven languages. This is a methodological case study of evidential boundaries, not a validation of the matrix as a universal framework. Its heterogeneity separates replicated findings from leads that still require confirmation.

The central result is that high probe F1 produces different evidential outcomes. The paired boolean comparison supports a bounded negative. The class or structure intervention fails its gate at one site and layer. Role-conditioned iterator renaming produces near-perfect embedding-layer scores, revealing that a nominal control can introduce a new shortcut. Legacy accumulator and index effects remain motivating observations. A single accuracy number, or a checkmark indicating that a control ran, cannot identify among these explanations.

## 2 Study design and evidence matrix

**Targets, models, and languages.** We treat five identifier properties and roles as related case studies, not measurement-equivalent instances of one construct. Accumulators update within loops. Indices or keys appear as counters or subscripts. Iterators are loop-bound names and their uses. Boolean labels distinguish assignment, conditional, loop, and return occurrences. Class or structure labels mark declared type names and references. Binary token probes label all other tokens as negative. Language-specific AST and regular-expression extractors lack a common multilingual gold audit. Protocols also differ. Accumulator and index use legacy token splits and test-maximized layers. Boolean mean-pools occurrences and uses problem groups. Iterator and class or structure use token labels, problem groups, validation-selected layers, and five seeds. XLCoST contains C, C++, C#, Java, JavaScript, PHP, and Python. Table 1 states coverage. Scores across rows do not estimate one common task.

**Four comparison questions.** Renaming asks whether lexical identity carries the prediction. Random-noun, single-character, and numeric strategies are role-agnostic textual substitutions. The misleading strategy is not role-agnostic. Target identifiers receive counter-role names while other identifiers receive role-looking names. The vocabulary therefore encodes the label. The all-same strategy is also non-bijective. Because every condition refits its probe and layer, neither strategy

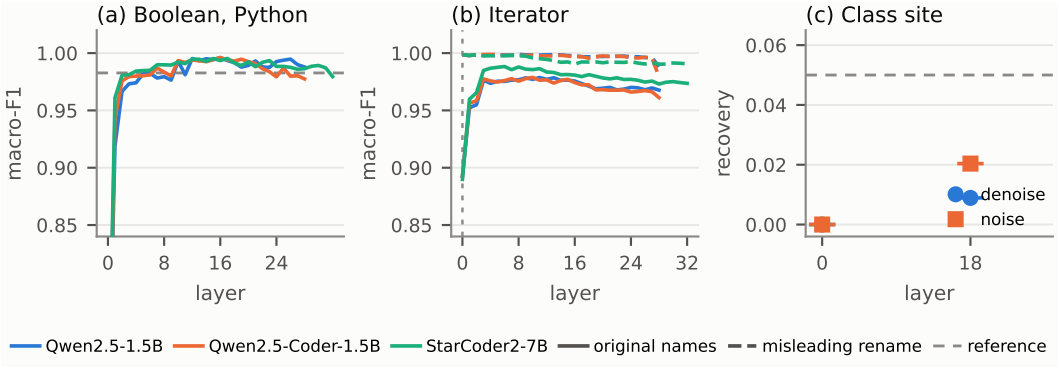


Figure 1: Three high probe scores with different comparisons. Boolean probes on Python stay near a masked source-line baseline across layers. Iterator probes saturate at embedding layer zero under role-conditioned renaming. Class-site residual patching remains below the $0.05$ recovery gate.

tests semantic robustness or a fixed invariant direction. We retain them as diagnostics of a failed intervention. A model-free character classifier with the target name masked asks whether local source text already determines the label. Randomized-label controls ask whether the probe family can fit a specified null. Boolean assigns a random label per variable name. Iterator and class or structure permute labels within programs. Full-residual patching asks whether replacing the complete site state moves a behavioral logit readout [Meng et al., 2022]. Its interpretation depends on the corruption, metric, and readout, so these choices belong in the evidence record [Zhang and Nanda, 2023, Heimersheim and Nanda, 2024].

These comparisons are not interchangeable. Role-agnostic refit renaming can measure recoverability after a cue changes, but role-conditioned renaming can create a cue. Positive selectivity can reject its specified permutation task while leaving a surface shortcut intact. A model-free comparator can establish surface sufficiency but cannot show how the model uses the feature. Full-residual patching tests site-state relevance, not use of the decoded probe direction, and only at its specified layer, span, direction, and readout.

**Evidence grades.** Boolean, iterator, and class or structure experiments use problem groups, validation-selected layers, five seeds, and BCa or clustered uncertainty. Iterator additionally reports a within-program label-permutation control and a name-only comparator, but their difference is not a paired clustered interval. Accumulator and index use token splits, test-maximized layers, and whole-word renaming. We retain their opposite effects as legacy hypotheses, not population estimates. A DeepSeek index run is excluded because a tokenizer offset error invalidates its activation alignment.

# 3 The same probe score supports different claims

## 3.1 Renaming diagnoses an intervention confound

Legacy Qwen2.5-1.5B runs first motivated the comparison. The index probe falls from $0.959$ to $0.687$ under misleading names, a change of $-0.272$, while the accumulator probe rises from $0.893$ to $0.972$, a change of $+0.079$. Token-level splitting, test-maximized layers, role-conditioned replacements, and refitting prevent a robustness interpretation. These are exploratory cue responses rather than a confirmed cross-role dissociation.

The modern iterator rerun reveals the intervention problem directly. Across the three models, baseline F1 is $0.976$, $0.978$, and $0.987$. The widest BCa $95\%$ interval is $[0.964, 0.982]$, and permutation selectivity is $+0.526$ to $+0.552$. Role-conditioned misleading names change F1 by $+0.023$, $+0.020$, and $+0.012$, but all three probes select embedding layer zero and reach $0.998$–$0.999$ (Figure 1b). The name pools encode target membership, so this result is compatible with relearning an inverted lexical rule. It does not support semantic robustness. Single-character and numeric names reduce F1 by $0.046$–$0.072$. A name-only classifier on the same $427$ Python programs reaches $0.918$, leaving an unpaired gap of $0.058$–$0.069$. Python-trained transfer ranges are $0.934$–$0.981$, $0.898$–$0.923$, and

$0.887$–$0.938$. Because target evaluation is not separated by one global cross-language problem partition, these scores establish descriptive transfer rather than language abstraction. Exploratory iterator patching appears in the appendix and is not the class or structure causal gate.

### 3.2 A model-free baseline bounds the boolean result

Boolean occurrence type provides the clearest bounded negative result. On a frozen Python sample, the three models reach $0.982$, $0.981$, and $0.988$ macro-F1. Selectivity ranges from $+0.272$ to $+0.288$. These values reject the specific hypothesis that an equally expressive probe predicts labels randomly assigned per variable name as well as it predicts the real usage labels. Yet a character classifier on the enclosing source line, with the variable name masked, reaches $0.983$ (Figure 1a). The probe-minus-surface differences are $-0.001$, $-0.002$, and $+0.005$.

The original exports retained no aligned predictions, so we reran both predictors on the frozen sample and used aligned occurrence identifiers. Pooling the five seed test folds gives $2{,}067$ seed-level predictions for $1{,}410$ unique occurrences in $915$ of the $1{,}301$ source problems, and predictor overlap is complete on every fold. The committed artifact retains the aggregate paired summary, not those prediction-level inputs. The probe-minus-surface estimates and problem-clustered $95\%$ intervals are $-0.002$ $[-0.017, 0.014]$, $-0.003$ $[-0.037, 0.014]$, and $+0.002$ $[-0.016, 0.017]$. Every interval covers zero and excludes a probe advantage larger than $0.014$–$0.017$ macro-F1. The masked-line comparator was the best of the reported surface feature families on this frozen sample, so the bound is conditional on that non-nested selection. That selection matters. Against the next-best Python comparator, the masked enclosing statement at $0.970$, the differences are $+0.010$ to $+0.017$. Across the four committed languages, probe minus best baseline ranges from $-0.004$ to $+0.021$, the largest value being PHP with StarCoder2-7B, and line masking is degenerate outside Python because detokenized XLCoST programs there occupy a single line. Five refit renaming conditions change F1 by at most $0.013$, but the role-conditioned and non-bijective strategies prevent a robustness claim. The supported conclusion is narrower. On Python, the probe does not outperform the selected masked local-syntax comparator by more than $0.014$–$0.017$. The other languages have unpaired differences only, so they carry no bound.

### 3.3 A site-state intervention fails its specified gate

The class or structure experiments and the boolean experiments use different controls. Across all three models, baseline F1 is $0.979$–$0.982$ and within-program permutation selectivity is $+0.503$–$+0.553$. Misleading class-style names change F1 by $-0.004$, $-0.004$, and $+0.004$, but this role-conditioned vocabulary prevents a robustness interpretation. Python-trained probes transfer to C++, JavaScript, and C at $0.885$–$0.980$. These results support decodability and performance above the specified within-program permutation task. They do not establish abstraction or a fixed rename-invariant direction.

We then test site-state relevance in Qwen2.5-1.5B by patching the full residual state. The experiment contains $288$ matched class-versus-function prompts in $48$ template clusters. Let $D = \ell(\texttt{True}) - \ell(\texttt{False})$. Denoising patches the class-source residual into a function prompt and defines $e_d = D_{\text{patched function}} - D_{\text{function}}$. Noising reverses the patch and defines $e_n = D_{\text{class}} - D_{\text{patched class}}$. Recovery is the ratio of the mean signed effect to the mean matched gap $D_{\text{class}} - D_{\text{function}}$. Percentile intervals resample template clusters. The gate requires positive, control-separated effects, recovery of at least $0.05$, and mean effects of at least $0.10$ in both directions. These thresholds are design choices rather than calibrated behavioral minima.

Both prompt classes prefer `True`, which limits the behavioral readout's effective decision range. Their mean logit differences are $+2.75$ for class prompts and $+1.78$ for function prompts, although the matched class-minus-function gap is $+0.97$ and has the expected sign for $98.6\%$ of pairs. The gate therefore tests whether patching transfers this relative distinction, not whether it flips a well-calibrated binary decision.

Denoising produces a mean effect of $0.0087$ with a $95\%$ interval $[-0.0007, 0.0188]$ and recovery $0.0089$. Noising produces $0.0199$ $[0.0109, 0.0296]$ with recovery $0.0204$. The full-residual intervention fails the specified gate (Figure 1c). The protocol and results entered the repository together, so we do not claim auditable preregistration. Recovery divides by the matched gap, so a compressed decision range does not explain a ratio of $0.009$ to $0.020$. We read this as a bounded negative at

Table 2: Outcome-aware evidence records. Each next test can change the stated claim.

| Case | Comparison and outcome | Matching and uncertainty | Claim and falsifier |
|---|---|---|---|
| Legacy roles | Refit accumulator $+0.079$, index $-0.272$ | Token split, test-selected layer, no interval | Exploratory only. Rerun grouped with label-blind renames |
| Iterator | Role-conditioned rename reaches L0 at 0.998–0.999 | Problem split and marginal BCa intervals | Control is confounded. Use bijective role-agnostic renames |
| Boolean | Probe minus masked line is $-0.003$ to $+0.002$ | 915 clusters with paired 95% intervals | Not supported on Python. No selected-baseline advantage above 0.017. Nest selection and add languages |
| Class/function | Bidirectional residual-patch gate fails | 48 template clusters with 95% intervals | Bounded effect at one site and layer. Test other spans and layers |

the tested site, layer, and span rather than evidence that class information is absent or unused. The intervals exclude mean effects above $0.0188$ and $0.0296$ logit-difference units in the tested directions. Decodability and selectivity therefore do not establish a substantive site-state contribution under this readout.

## 4 An outcome-aware reporting matrix

The contribution is a reporting obligation, not a new control or evidence ladder. Each record states the target and prediction unit, claim and estimand, comparator and matching rule, uncertainty, outcome, and falsifying next test. We distinguish five claim types. Decodability exceeds a label baseline. Capacity control exceeds a specified randomized-label task. Cue sensitivity uses an intervention that does not itself encode the label. Surface sufficiency compares a model-free predictor on a matched population. Site-state causal relevance requires a gate-specified intervention and behavioral readout. No claim entails another.

These heterogeneous cases demonstrate how the record changes interpretation. They do not validate the matrix as universal. A claim of cross-language abstraction remains open until one extractor, global problem partition, and matched control battery cover a complete language grid. Replication packages should publish the record beside metric tables and state the smallest effect the design can exclude.

## 5 Conclusion

High probe accuracy is not a common unit of interpretability evidence. In these cases, matched surface prediction supports a bounded negative, role-conditioned renaming exposes a shortcut, and a specified patching gate fails at one site and layer. Publishing each claim as an outcome-aware record lets null and confounded outcomes constrain conclusions rather than disappear behind a control checkmark.

## 6 Limitations

The study collectively covers three models, five roles, and seven languages, but not a complete $3 \times 5 \times 7$ design, and Table 1 gives the per-target coverage. The targets are heterogeneous, and their language-specific extractors lack a common gold audit. Role-conditioned and all-same renames can encode or alter labels. Legacy accumulator-index results lack problem groups and scope-aware renaming. Iterator has no paired probe-minus-surface interval, and its target evaluation does not enforce one global cross-language problem partition. The boolean bound is conditional on a surface comparator selected on the frozen sample, its sign varies across languages, and its prediction-level paired inputs are not retained. Full-residual patching covers one model, layer, site, and readout. The gate and stopping rule were not registered in advance. These cases motivate the reporting matrix rather than validate it.

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# A Per-role control results

Table 3: Complete boolean occurrence-type probe and model-free baseline results. The best baseline is selected from name-only, masked statement, masked line, masked window, covariate-only, and majority features. $\rho$ is a descriptive one-instance score step, not an interval or equivalence test. Aligned probe/baseline predictions were not retained.

| language | model | problems | probe F1 | best baseline | difference | $\rho$ | selectivity |
|---|---|---|---|---|---|---|---|
| Java | Qwen2.5-1.5B | 1,427 | 0.982 | 0.980 | +0.002 | 0.0120 | +0.255 |
| Java | Qwen2.5-Coder-1.5B | 1,427 | 0.976 | 0.980 | -0.004 | 0.0079 | +0.271 |
| Java | StarCoder2-7B | 1,427 | 0.990 | 0.980 | +0.010 | 0.0120 | +0.243 |
| JavaScript | Qwen2.5-1.5B | 1,351 | 0.989 | 0.992 | -0.003 | 0.0170 | +0.132 |
| JavaScript | Qwen2.5-Coder-1.5B | 1,351 | 0.998 | 0.992 | +0.006 | 0.0170 | +0.150 |
| JavaScript | StarCoder2-7B | 1,351 | 0.988 | 0.992 | -0.004 | 0.0170 | +0.120 |
| PHP | Qwen2.5-1.5B | 1,305 | 0.983 | 0.969 | +0.014 | 0.0080 | +0.214 |
| PHP | Qwen2.5-Coder-1.5B | 1,305 | 0.975 | 0.969 | +0.006 | 0.0080 | +0.216 |
| PHP | StarCoder2-7B | 1,305 | 0.990 | 0.969 | +0.021 | 0.0080 | +0.240 |
| Python | Qwen2.5-1.5B | 1,301 | 0.982 | 0.983 | -0.001 | 0.0068 | +0.280 |
| Python | Qwen2.5-Coder-1.5B | 1,301 | 0.981 | 0.983 | -0.002 | 0.0068 | +0.272 |
| Python | StarCoder2-7B | 1,301 | 0.988 | 0.983 | +0.005 | 0.0068 | +0.288 |

Table 4: Boolean Python identifier-renaming audit. C1–C5 are the five committed renaming conditions. Each condition refits a probe and reselects its layer, so deltas measure recoverability rather than fixed-direction invariance. Intervals are problem-clustered. Empty renaming fields for other languages indicate that those runs were not committed, not a null result.

| model | C1 | C2 | C3 | C4 | C5 |
|---|---|---|---|---|---|
| Qwen2.5-1.5B | +0.003 | -0.004 | +0.001 | -0.003 | +0.001 |
| Qwen2.5-Coder-1.5B | +0.001 | -0.010 | +0.006 | -0.004 | -0.006 |
| StarCoder2-7B | -0.003 | -0.013 | -0.006 | -0.001 | -0.006 |

**Renaming diagnostic.** The misleading strategy uses role-conditioned vocabularies. Target and other identifiers receive names from different pools, so a refit probe can recover the label lexically. The all-same strategy is non-bijective and can alter label extraction. These rows diagnose intervention confounds rather than semantic robustness.

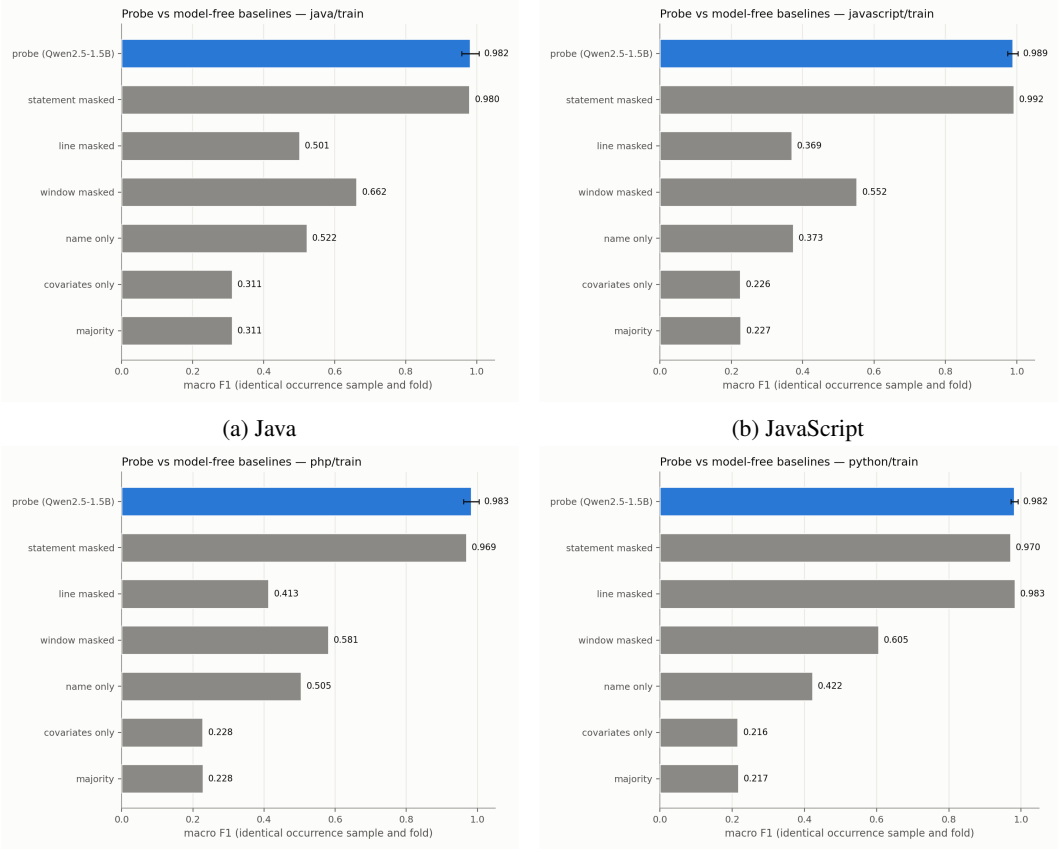
Table 5: Complete model-free boolean baselines by language. Baselines are shared across the three neural models because they use the same frozen examples and surface features.

| language | majority | name | statement | line | window | covariates |
|---|---|---|---|---|---|---|
| Java | 0.311 | 0.522 | 0.980 | 0.501 | 0.662 | 0.311 |
| JavaScript | 0.227 | 0.373 | 0.992 | 0.369 | 0.552 | 0.226 |
| PHP | 0.228 | 0.505 | 0.969 | 0.413 | 0.581 | 0.228 |
| Python | 0.217 | 0.422 | 0.970 | 0.983 | 0.605 | 0.216 |

**Boolean probe figures.** The baseline panels retain the individual surface comparators hidden by the best-baseline table. Renaming is available only for Python. The absence of corresponding figures in other languages is a coverage hole.

Table 6: Complete class/structure perturbation results under the shared five-seed protocol. Each condition refits a probe and selects its own layer on validation data, so cross-condition deltas do not test a fixed direction.

| model | strategy | layer | F1 | 95% CI | control F1 | selectivity | Δ |
|---|---|---|---|---|---|---|---|
| Qwen2.5-1.5B | baseline | L18 | 0.979 | [0.967, 0.987] | 0.427 | +0.552 | N/A |
| Qwen2.5-1.5B | random_nouns | L20 | 0.986 | [0.976, 0.992] | 0.426 | +0.560 | +0.007 |
| Qwen2.5-1.5B | single_chars | L11 | 0.957 | [0.936, 0.969] | 0.423 | +0.534 | -0.022 |
| Qwen2.5-1.5B | all_same | L0 | 1.000 | [1.000, 1.000] | 0.440 | +0.560 | +0.021 |
| Qwen2.5-1.5B | numeric_vars | L21 | 0.965 | [0.952, 0.972] | 0.422 | +0.543 | -0.014 |
| Qwen2.5-1.5B | misleading_class_struct | L7 | 0.976 | [0.962, 0.982] | 0.416 | +0.559 | -0.004 |
| Qwen2.5-Coder-1.5B | baseline | L8 | 0.982 | [0.969, 0.989] | 0.429 | +0.553 | N/A |
| Qwen2.5-Coder-1.5B | random_nouns | L19 | 0.978 | [0.965, 0.985] | 0.427 | +0.551 | -0.004 |
| Qwen2.5-Coder-1.5B | single_chars | L7 | 0.956 | [0.936, 0.967] | 0.423 | +0.533 | -0.026 |
| Qwen2.5-Coder-1.5B | all_same | L0 | 1.000 | [1.000, 1.000] | 0.440 | +0.560 | +0.018 |
| Qwen2.5-Coder-1.5B | numeric_vars | L21 | 0.968 | [0.955, 0.975] | 0.424 | +0.544 | -0.014 |
| Qwen2.5-Coder-1.5B | misleading_class_struct | L7 | 0.978 | [0.967, 0.984] | 0.415 | +0.562 | -0.004 |
| StarCoder2-7B | baseline | L5 | 0.981 | [0.967, 0.990] | 0.478 | +0.503 | N/A |
| StarCoder2-7B | random_nouns | L6 | 0.976 | [0.962, 0.983] | 0.466 | +0.510 | -0.006 |
| StarCoder2-7B | single_chars | L4 | 0.957 | [0.921, 0.971] | 0.475 | +0.482 | -0.025 |
| StarCoder2-7B | all_same | L0 | 1.000 | [1.000, 1.000] | 0.445 | +0.555 | +0.019 |
| StarCoder2-7B | numeric_vars | L23 | 0.968 | [0.956, 0.975] | 0.451 | +0.517 | -0.013 |
| StarCoder2-7B | misleading_class_struct | L5 | 0.985 | [0.976, 0.990] | 0.467 | +0.518 | +0.004 |




(a) Java

(b) JavaScript

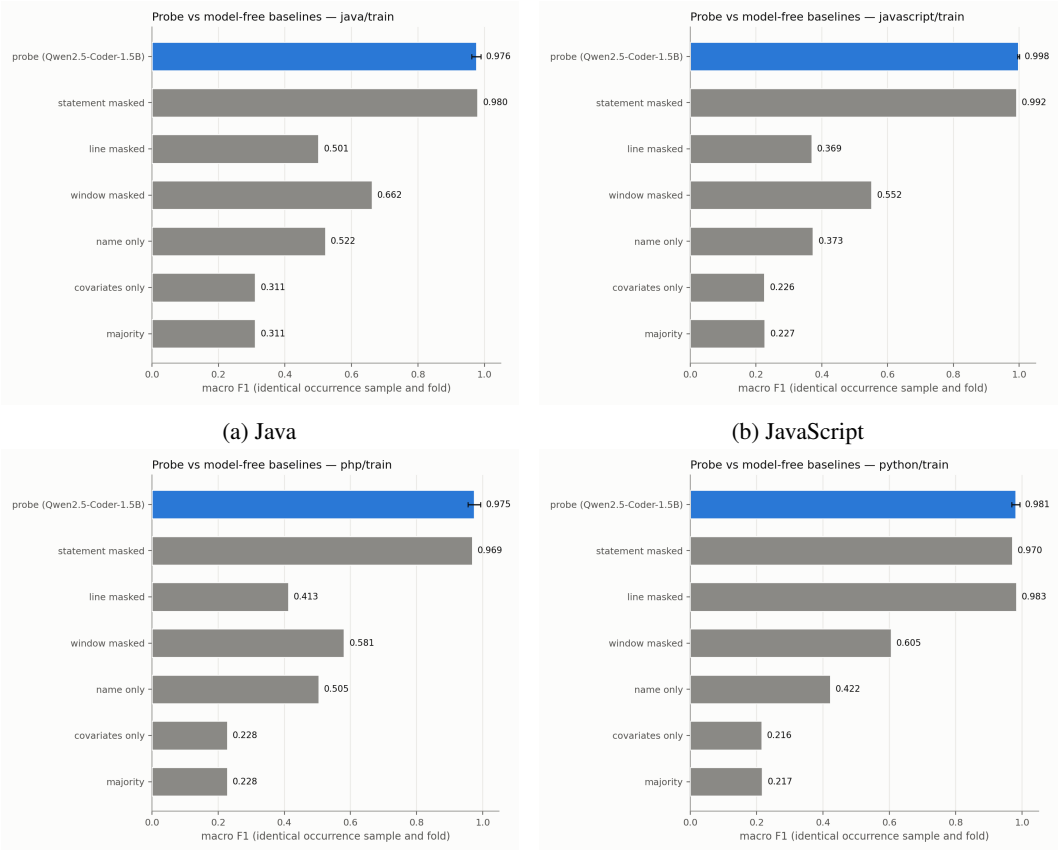
(c) PHP

(d) Python

Figure 2: Boolean probe and model-free baselines for Qwen2.5-1.5B across the four committed languages.

Table 7: Class/structure cross-language results. Transfer uses the Python-selected layer without a shared source-target problem partition. In-domain layers are selected separately. Only languages present in the committed exports are shown.

| model | language | programs | in-domain layer | in-domain F1 | transfer accuracy | transfer F1 |
|---|---|---|---|---|---|---|
| Qwen2.5-1.5B | Python | 400 | L18 | 0.979 | N/A | N/A |
| Qwen2.5-1.5B | C++ | 606 | L10 | 0.991 | 0.993 | 0.933 |
| Qwen2.5-1.5B | JavaScript | 285 | L5 | 0.987 | 0.998 | 0.957 |
| Qwen2.5-1.5B | C | 97 | L8 | 0.986 | 0.984 | 0.886 |
| Qwen2.5-Coder-1.5B | Python | 400 | L8 | 0.982 | N/A | N/A |
| Qwen2.5-Coder-1.5B | C++ | 606 | L8 | 0.988 | 0.991 | 0.918 |
| Qwen2.5-Coder-1.5B | JavaScript | 285 | L6 | 0.985 | 0.996 | 0.913 |
| Qwen2.5-Coder-1.5B | C | 97 | L20 | 0.980 | 0.983 | 0.885 |
| StarCoder2-7B | Python | 400 | L5 | 0.981 | N/A | N/A |
| StarCoder2-7B | C++ | 606 | L5 | 0.992 | 0.996 | 0.965 |
| StarCoder2-7B | JavaScript | 285 | L3 | 0.989 | 0.999 | 0.980 |
| StarCoder2-7B | C | 97 | L5 | 0.986 | 0.987 | 0.906 |




(a) Java

(b) JavaScript

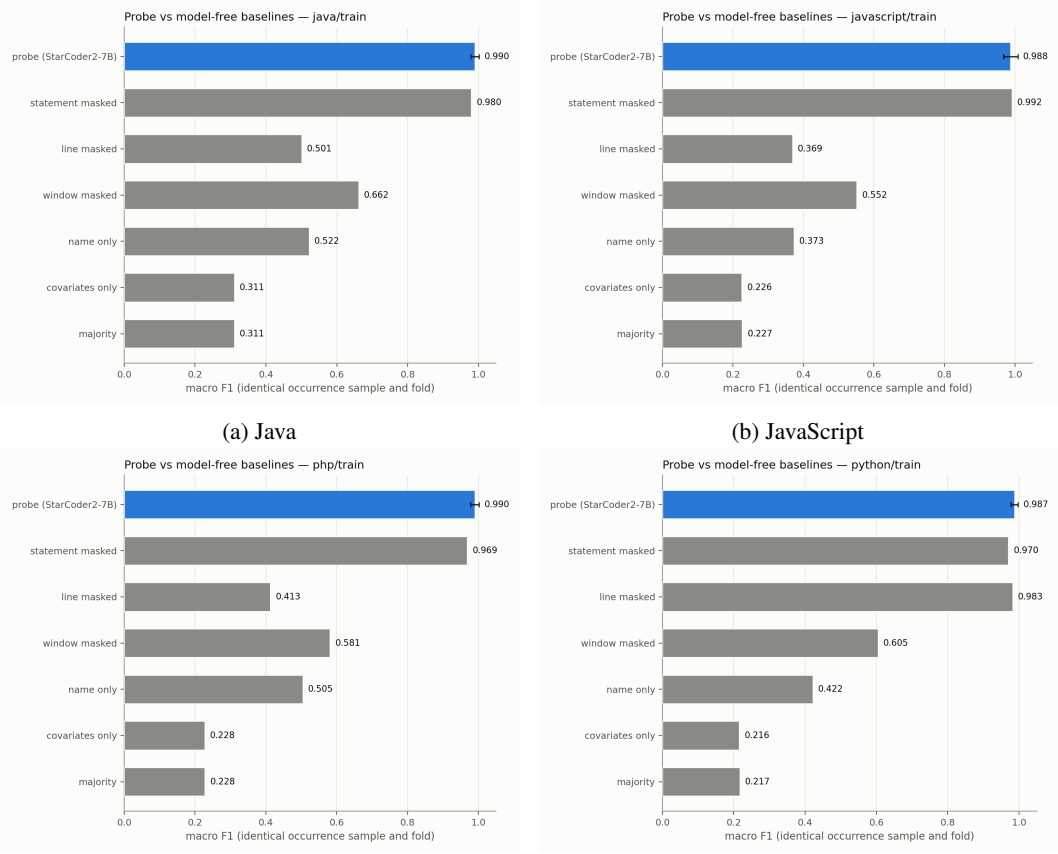
(c) PHP

(d) Python

Figure 3: Boolean probe and model-free baselines for Qwen2.5-Coder-1.5B across the four committed languages.

Table 8: Layerwise class/structure robustness summaries. The F1 range is taken over all committed layers. Maximum cosine similarity is measured against the baseline activation direction for each renamed condition.

| model | strategy | minimum F1 | maximum F1 (layer) | maximum cosine (layer) |
|---|---|---|---|---|
| Qwen2.5-1.5B | all_same | 0.999 | 1.000 (L0) | 0.105 (L11) |
| Qwen2.5-1.5B | baseline | 0.806 | 0.983 (L18) | N/A |
| Qwen2.5-1.5B | misleading_class_struct | 0.708 | 0.980 (L7) | 0.311 (L15) |
| Qwen2.5-1.5B | numeric_vars | 0.445 | 0.966 (L21) | 0.329 (L22) |
| Qwen2.5-1.5B | random_nouns | 0.529 | 0.989 (L15) | 0.493 (L26) |
| Qwen2.5-1.5B | single_chars | 0.530 | 0.965 (L7) | 0.392 (L15) |
| Qwen2.5-Coder-1.5B | all_same | 0.999 | 1.000 (L0) | 0.126 (L28) |
| Qwen2.5-Coder-1.5B | baseline | 0.807 | 0.985 (L7) | N/A |
| Qwen2.5-Coder-1.5B | misleading_class_struct | 0.708 | 0.978 (L7) | 0.324 (L7) |
| Qwen2.5-Coder-1.5B | numeric_vars | 0.445 | 0.969 (L24) | 0.352 (L22) |
| Qwen2.5-Coder-1.5B | random_nouns | 0.530 | 0.983 (L23) | 0.464 (L19) |
| Qwen2.5-Coder-1.5B | single_chars | 0.530 | 0.959 (L7) | 0.415 (L15) |
| StarCoder2-7B | all_same | 0.999 | 1.000 (L0) | 0.101 (L21) |
| StarCoder2-7B | baseline | 0.763 | 0.984 (L6) | N/A |
| StarCoder2-7B | misleading_class_struct | 0.717 | 0.985 (L3) | 0.387 (L3) |
| StarCoder2-7B | numeric_vars | 0.443 | 0.971 (L26) | 0.369 (L22) |
| StarCoder2-7B | random_nouns | 0.502 | 0.979 (L8) | 0.511 (L3) |
| StarCoder2-7B | single_chars | 0.529 | 0.962 (L3) | 0.467 (L6) |




(a) Java

(b) JavaScript

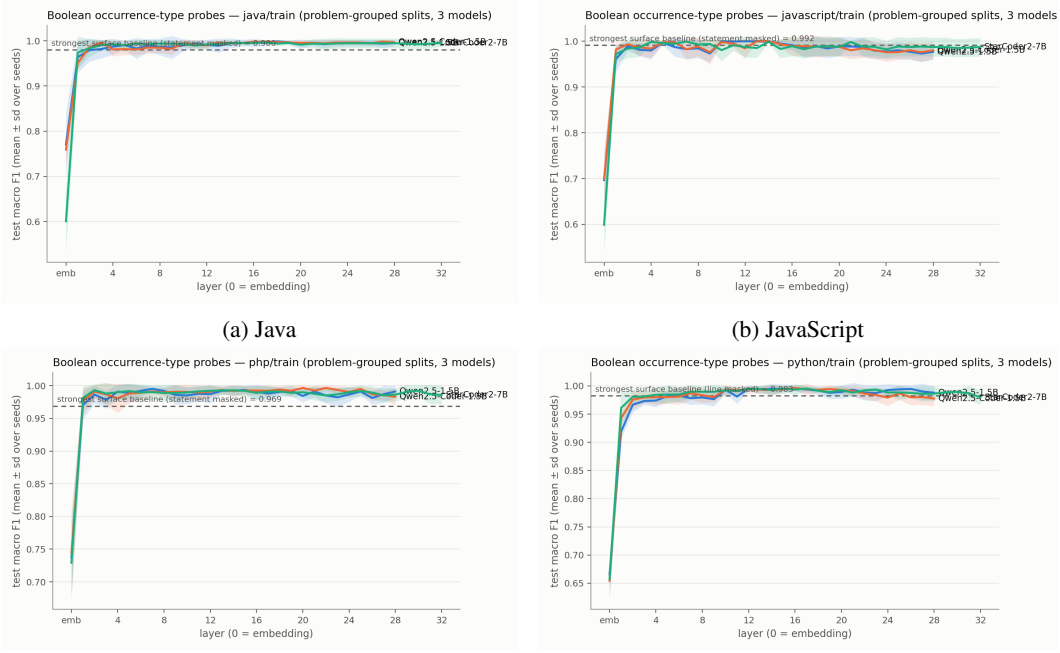
(c) PHP

(d) Python

Figure 4: Boolean probe and model-free baselines for StarCoder2-7B across the four committed languages.

Table 9: Complete iterator perturbation results under the shared five-seed protocol. Each condition refits a probe and selects its own layer on validation data. Role-conditioned and non-bijective strategies are confounded diagnostics rather than robustness tests.

| model | strategy | layer | F1 | 95% CI | control F1 | selectivity | Δ |
|---|---|---|---|---|---|---|---|
| Qwen2.5-1.5B | baseline | L7 | 0.976 | [0.964, 0.982] | 0.424 | +0.552 | N/A |
| Qwen2.5-1.5B | random_nouns | L11 | 0.961 | [0.951, 0.967] | 0.423 | +0.538 | -0.015 |
| Qwen2.5-1.5B | single_chars | L9 | 0.906 | [0.896, 0.917] | 0.428 | +0.479 | -0.069 |
| Qwen2.5-1.5B | all_same | L1 | 1.000 | [1.000, 1.000] | 0.456 | +0.544 | +0.024 |
| Qwen2.5-1.5B | numeric_vars | L23 | 0.910 | [0.899, 0.920] | 0.440 | +0.471 | -0.065 |
| Qwen2.5-1.5B | misleading_iterator | L0 | 0.999 | [0.998, 0.999] | 0.403 | +0.595 | +0.023 |
| Qwen2.5-Coder-1.5B | baseline | L7 | 0.978 | [0.968, 0.983] | 0.429 | +0.549 | N/A |
| Qwen2.5-Coder-1.5B | random_nouns | L12 | 0.956 | [0.946, 0.962] | 0.423 | +0.533 | -0.022 |
| Qwen2.5-Coder-1.5B | single_chars | L8 | 0.906 | [0.895, 0.916] | 0.426 | +0.480 | -0.072 |
| Qwen2.5-Coder-1.5B | all_same | L1 | 1.000 | [1.000, 1.000] | 0.459 | +0.541 | +0.022 |
| Qwen2.5-Coder-1.5B | numeric_vars | L22 | 0.911 | [0.896, 0.921] | 0.440 | +0.470 | -0.068 |
| Qwen2.5-Coder-1.5B | misleading_iterator | L0 | 0.998 | [0.997, 0.999] | 0.402 | +0.596 | +0.020 |
| StarCoder2-7B | baseline | L11 | 0.987 | [0.981, 0.990] | 0.461 | +0.526 | N/A |
| StarCoder2-7B | random_nouns | L6 | 0.975 | [0.967, 0.979] | 0.455 | +0.520 | -0.012 |
| StarCoder2-7B | single_chars | L6 | 0.941 | [0.933, 0.949] | 0.458 | +0.483 | -0.046 |
| StarCoder2-7B | all_same | L2 | 1.000 | [1.000, 1.000] | 0.460 | +0.540 | +0.013 |
| StarCoder2-7B | numeric_vars | L26 | 0.934 | [0.921, 0.941] | 0.454 | +0.480 | -0.053 |
| StarCoder2-7B | misleading_iterator | L0 | 0.998 | [0.997, 0.999] | 0.406 | +0.592 | +0.012 |




(a) Java (b) JavaScript

(c) PHP (d) Python

Figure 5: Boolean probe layer curves per language across the three models.

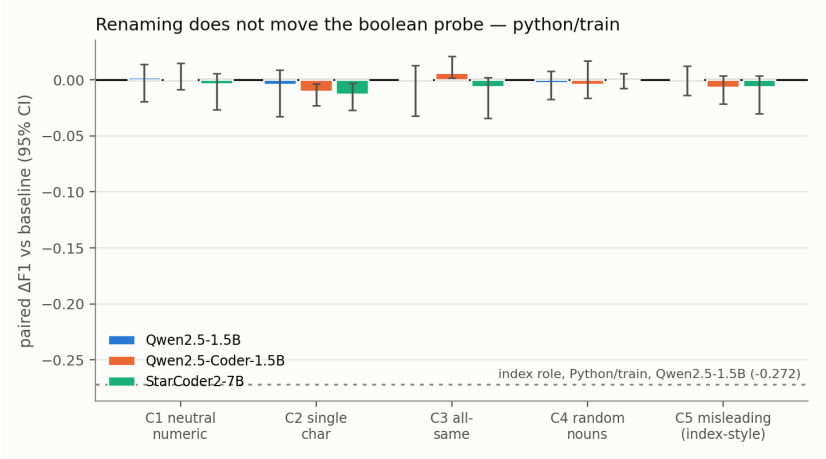



Figure 6: Problem-paired boolean refit-performance deltas on Python. No corresponding committed renaming run exists for other languages.

Table 10: Model-free iterator surface baselines on the same 427 Python programs. Scores are occurrence-level macro-F1 and are not a paired probe-minus-surface interval.

| comparator | macro-F1 |
|---|---|
| majority | 0.363 |
| covariates | 0.361 |
| masked line | 0.844 |
| masked statement | 0.842 |
| masked window | 0.869 |
| name only | 0.918 |

Table 11: Complete iterator cross-language transfer results. Transfer columns use the Python-selected layer.

| model | language | programs | in-domain layer | in-domain F1 | transfer accuracy | transfer F1 |
|---|---|---|---|---|---|---|
| Qwen2.5-1.5B | Python | 256 | L9 | 0.971 | N/A | N/A |
| Qwen2.5-1.5B | Java | 188 | L13 | 0.990 | 0.993 | 0.956 |
| Qwen2.5-1.5B | C++ | 191 | L12 | 0.992 | 0.994 | 0.965 |
| Qwen2.5-1.5B | C | 191 | L7 | 0.972 | 0.987 | 0.934 |
| Qwen2.5-1.5B | C# | 189 | L13 | 0.987 | 0.993 | 0.959 |
| Qwen2.5-1.5B | JavaScript | 266 | L9 | 0.985 | 0.992 | 0.966 |
| Qwen2.5-1.5B | PHP | 253 | L12 | 0.986 | 0.996 | 0.981 |
| Qwen2.5-Coder-1.5B | Python | 256 | L10 | 0.978 | N/A | N/A |
| Qwen2.5-Coder-1.5B | Java | 188 | L11 | 0.989 | 0.986 | 0.908 |
| Qwen2.5-Coder-1.5B | C++ | 191 | L12 | 0.988 | 0.988 | 0.923 |
| Qwen2.5-Coder-1.5B | C | 191 | L6 | 0.967 | 0.982 | 0.898 |
| Qwen2.5-Coder-1.5B | C# | 189 | L14 | 0.984 | 0.985 | 0.909 |
| Qwen2.5-Coder-1.5B | JavaScript | 266 | L10 | 0.982 | 0.983 | 0.915 |
| Qwen2.5-Coder-1.5B | PHP | 253 | L12 | 0.988 | 0.986 | 0.923 |
| StarCoder2-7B | Python | 256 | L9 | 0.986 | N/A | N/A |
| StarCoder2-7B | Java | 188 | L7 | 0.992 | 0.980 | 0.887 |
| StarCoder2-7B | C++ | 191 | L6 | 0.989 | 0.984 | 0.906 |
| StarCoder2-7B | C | 191 | L10 | 0.971 | 0.981 | 0.901 |
| StarCoder2-7B | C# | 189 | L8 | 0.987 | 0.980 | 0.889 |
| StarCoder2-7B | JavaScript | 266 | L7 | 0.989 | 0.980 | 0.911 |
| StarCoder2-7B | PHP | 253 | L12 | 0.989 | 0.987 | 0.938 |

Table 12: Layerwise iterator robustness summaries. F1 ranges cover every committed layer. Cosine similarity compares each perturbation direction with the baseline direction.

| model | strategy | minimum F1 | maximum F1 (layer) | maximum cosine (layer) |
|---|---|---|---|---|
| Qwen2.5-1.5B | all_same | 0.999 | 1.000 (L0) | 0.200 (L15) |
| Qwen2.5-1.5B | baseline | 0.893 | 0.979 (L11) | N/A |
| Qwen2.5-1.5B | misleading_iterator | 0.988 | 0.999 (L8) | 0.129 (L7) |
| Qwen2.5-1.5B | numeric_vars | 0.550 | 0.913 (L23) | 0.255 (L19) |
| Qwen2.5-1.5B | random_nouns | 0.628 | 0.964 (L12) | 0.392 (L19) |
| Qwen2.5-1.5B | single_chars | 0.658 | 0.918 (L7) | 0.265 (L21) |
| Qwen2.5-Coder-1.5B | all_same | 0.999 | 1.000 (L0) | 0.227 (L18) |
| Qwen2.5-Coder-1.5B | baseline | 0.893 | 0.979 (L10) | N/A |
| Qwen2.5-Coder-1.5B | misleading_iterator | 0.979 | 0.999 (L4) | 0.179 (L20) |
| Qwen2.5-Coder-1.5B | numeric_vars | 0.550 | 0.913 (L23) | 0.277 (L19) |
| Qwen2.5-Coder-1.5B | random_nouns | 0.628 | 0.961 (L16) | 0.448 (L14) |
| Qwen2.5-Coder-1.5B | single_chars | 0.658 | 0.912 (L8) | 0.299 (L17) |
| StarCoder2-7B | all_same | 0.999 | 1.000 (L0) | 0.169 (L20) |
| StarCoder2-7B | baseline | 0.890 | 0.988 (L6) | N/A |
| StarCoder2-7B | misleading_iterator | 0.990 | 0.998 (L0) | 0.234 (L21) |
| StarCoder2-7B | numeric_vars | 0.549 | 0.937 (L24) | 0.271 (L25) |
| StarCoder2-7B | random_nouns | 0.603 | 0.975 (L5) | 0.385 (L10) |
| StarCoder2-7B | single_chars | 0.657 | 0.946 (L5) | 0.359 (L12) |

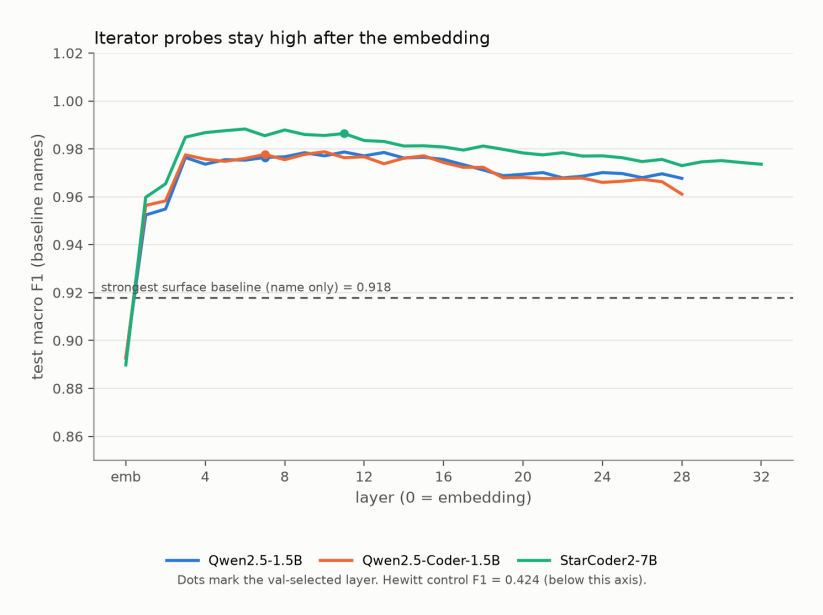



Figure 7: Iterator probe macro-F1 across layers on original names, with the name-only surface baseline marked.

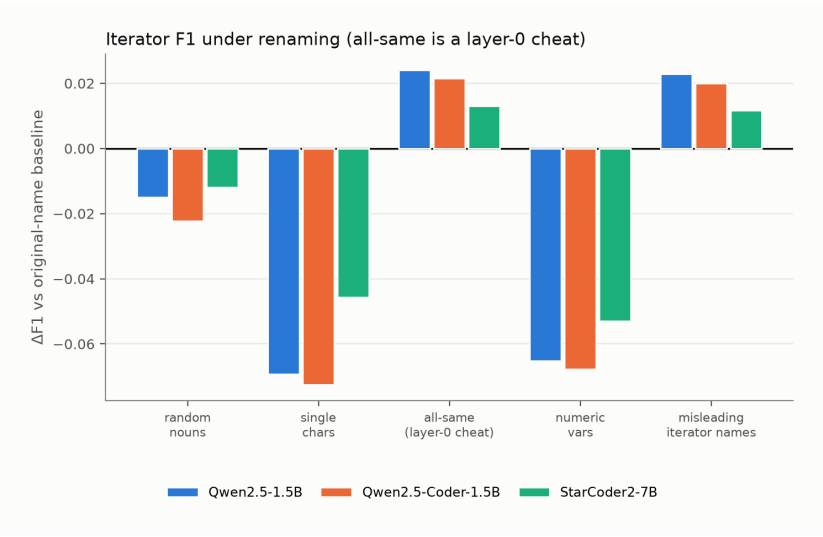



Figure 8: Iterator perturbation deltas relative to baseline.

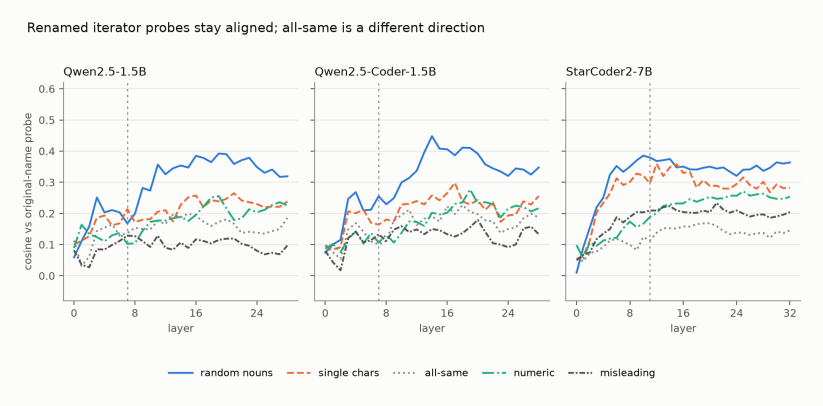



Figure 9: Iterator cosine similarity to the baseline direction.

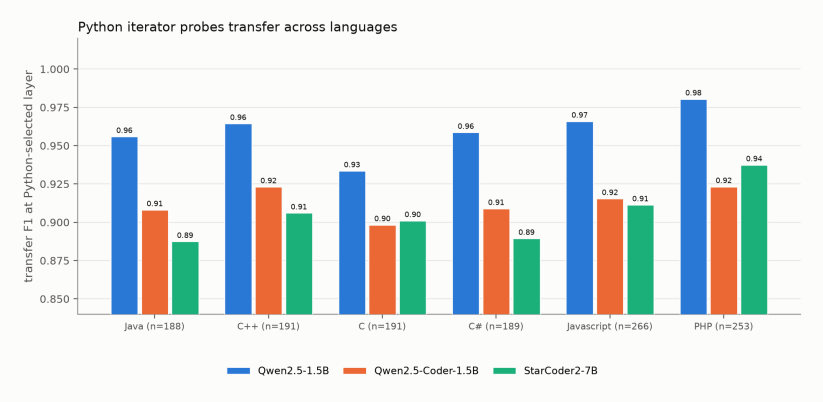



Figure 10: Available iterator cross-language transfer results.

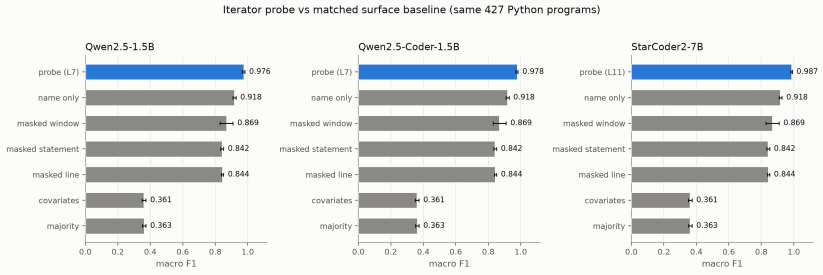



Figure 11: Iterator probe versus matched model-free surface baselines on the same 427 Python programs.

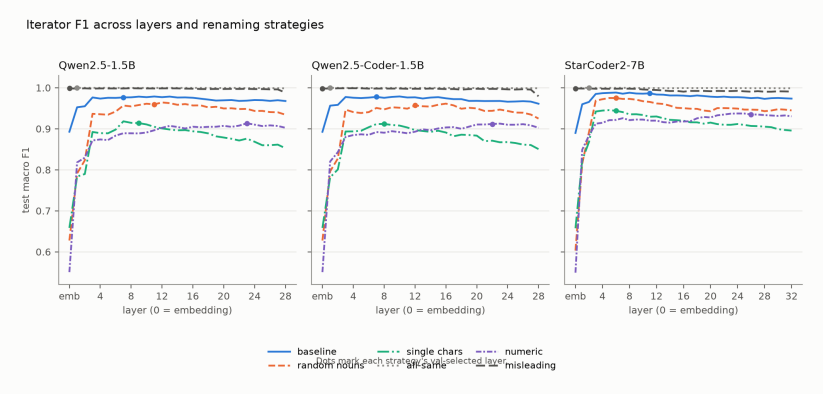



Figure 12: Iterator probe macro-F1 across layers and renaming strategies.

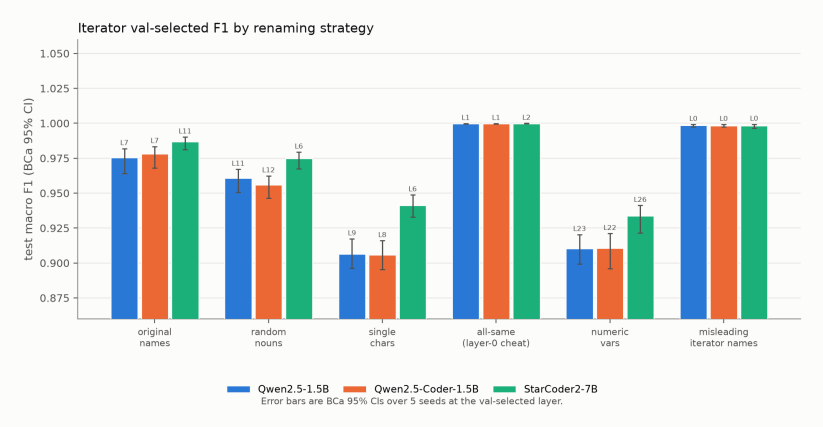



Figure 13: Iterator val-selected F1 by renaming strategy, with BCa 95% intervals.

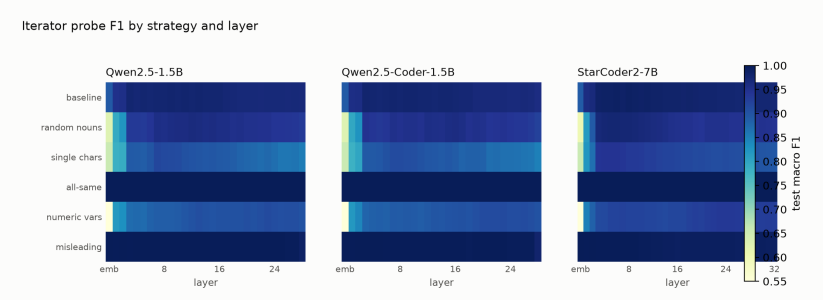



Figure 14: Iterator probe F1 by strategy and layer.

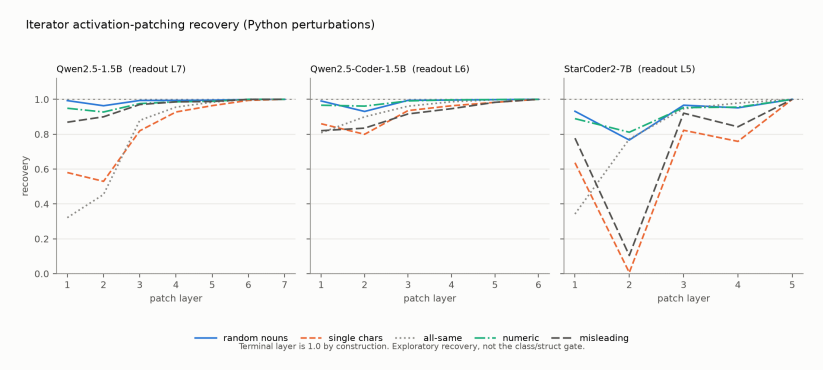



Figure 15: Exploratory iterator activation-patching recovery under Python name perturbations. The terminal layer is one by construction.

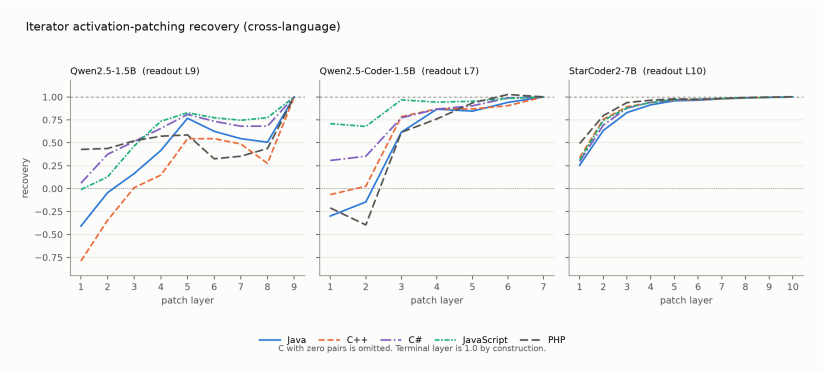



Figure 16: Exploratory iterator activation-patching recovery under cross-language transfer. C-language rows with zero pairs are omitted.

# B Causal intervention details

Table 13: Complete gate-specified class/structure patching summary in Qwen2.5-1.5B. The causal gate required mean effect at least 0.10, a positive interval, separation from controls, and recovery at least 0.05 in both directions.

| layer | span | direction | control | $n$ | mean effect | 95% CI | recovery | minus random |
|---|---|---|---|---|---|---|---|---|
| 0 | query_name | denoise | target | 288 | +0.0000 | [0.0000, 0.0000] | +0.0000 | N/A |
| 0 | query_name | noise | target | 288 | +0.0000 | [0.0000, 0.0000] | +0.0000 | N/A |
| 18 | placebo | denoise | target | 288 | +0.0015 | [-0.0010, 0.0043] | +0.0015 | N/A |
| 18 | placebo | noise | target | 288 | +0.0006 | [-0.0019, 0.0032] | +0.0006 | N/A |
| 18 | query_name | denoise | random | 288 | -0.0107 | [-0.0205, -0.0014] | -0.0110 | N/A |
| 18 | query_name | denoise | same_source | 288 | +0.0004 | [-0.0009, 0.0016] | +0.0004 | N/A |
| 18 | query_name | denoise | target | 288 | +0.0087 | [-0.0007, 0.0188] | +0.0089 | +0.0194 |
| 18 | query_name | noise | random | 288 | +0.0096 | [0.0023, 0.0164] | +0.0099 | N/A |
| 18 | query_name | noise | same_source | 288 | -0.0005 | [-0.0018, 0.0006] | -0.0006 | N/A |
| 18 | query_name | noise | target | 288 | +0.0199 | [0.0109, 0.0296] | +0.0204 | +0.0103 |

**Status.** The iterator recovery exports predate the gate-specified class/structure protocol. They report descriptive recovery curves without clustered confidence intervals or matched placebo/random controls. The terminal value is one by construction, and C-language cross-language rows with zero pairs are missing data rather than null effects.

Table 14: Aggregate probe-link diagnostic for the class/structure evaluation pairs. Absolute values are reported because pair orientation is arbitrary. Large probe movement alongside small behavioral movement is descriptive of the failed causal transfer, not evidence that probe movement causes behavior.

| quantity | mean absolute | median absolute |
|---|---|---|
| probe movement | 15.786 | 16.302 |
| behavioral movement | 0.038 | 0.031 |

Table 15: Class/structure gate and readout audit. Both prompt classes favor `True`. The pairwise gap remains separable, but the binary readout is not calibrated to distinguish function prompts.

| diagnostic | estimate | 95% CI | gate |
|---|---|---|---|
| class logit difference | 2.752 | [2.624, 2.898] | pass |
| function logit difference | 1.778 | [1.652, 1.909] | fail |
| class-minus-function gap | 0.974 | [0.908, 1.035] | pass |
| pair-gap accuracy | 0.986 | [0.962, 0.993] | pass |
| query/declaration probe AUC | 0.943/0.905 | N/A | pass |

Table 16: Patching audit trail. Completeness establishes that the scheduled Qwen evaluation finished. It does not turn the failed causal gate into positive evidence. The controller stopped before Coder and StarCoder2 after the Qwen null.

| audit item | value |
|---|---|
| behavior rows present/expected | 1152/1152 |
| primary rows present/expected | 4032/4032 |
| baseline probe-gap/behavior Spearman | 0.164 [0.023, 0.286] |
| controller terminal state | `stopped_qwen_null` |

Table 17: Exploratory iterator activation-patching recovery for Qwen2.5-1.5B, within-language perturbations.

| condition | readout | pairs | clean/corrupt | L1 | L2 | L3 | L4 | L5 | L6 | L7 |
|---|---|---|---|---|---|---|---|---|---|---|
| random_nouns | L7 | 150 | 0.999/0.307 | 0.993 | 0.963 | 0.994 | 0.994 | 0.995 | 0.999 | 1.000 |
| single_chars | L7 | 150 | 0.995/0.712 | 0.580 | 0.530 | 0.820 | 0.927 | 0.964 | 0.994 | 1.000 |
| all_same | L7 | 150 | 0.999/0.138 | 0.321 | 0.454 | 0.879 | 0.955 | 0.984 | 1.000 | 1.000 |
| numeric_vars | L7 | 150 | 0.999/0.211 | 0.949 | 0.927 | 0.976 | 0.986 | 0.989 | 1.000 | 1.000 |
| misleading_iterator | L7 | 150 | 0.998/0.324 | 0.869 | 0.900 | 0.970 | 0.985 | 0.988 | 1.000 | 1.000 |

Table 18: Exploratory iterator activation-patching recovery for Qwen2.5-1.5B, cross-language transfer.

| condition | readout | pairs | clean/corrupt | L1 | L2 | L3 | L4 | L5 | L6 | L7 | L8 | L9 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Java | L9 | 20 | 0.998/0.768 | -0.409 | -0.044 | 0.165 | 0.416 | 0.767 | 0.625 | 0.544 | 0.505 | 1.000 |
| C++ | L9 | 20 | 0.996/0.801 | -0.790 | -0.345 | 0.010 | 0.148 | 0.544 | 0.545 | 0.486 | 0.275 | 1.000 |
| C | L9 | 0 | 0.000/0.000 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A |
| C# | L9 | 30 | 0.993/0.801 | 0.059 | 0.374 | 0.514 | 0.657 | 0.809 | 0.734 | 0.680 | 0.681 | 1.000 |
| JavaScript | L9 | 44 | 0.996/0.828 | -0.011 | 0.128 | 0.465 | 0.734 | 0.827 | 0.773 | 0.745 | 0.774 | 1.000 |
| PHP | L9 | 12 | 0.987/0.810 | 0.427 | 0.437 | 0.521 | 0.571 | 0.586 | 0.325 | 0.353 | 0.437 | 1.000 |

Table 19: Exploratory iterator activation-patching recovery for Qwen2.5-Coder-1.5B, within-language perturbations.

| condition | readout | pairs | clean/corrupt | L1 | L2 | L3 | L4 | L5 | L6 |
|---|---|---|---|---|---|---|---|---|---|
| random_nouns | L6 | 150 | 0.999/0.138 | 0.991 | 0.930 | 0.994 | 0.996 | 0.998 | 1.000 |
| single_chars | L6 | 150 | 0.996/0.614 | 0.860 | 0.799 | 0.935 | 0.963 | 0.983 | 1.000 |
| all_same | L6 | 150 | 0.999/0.142 | 0.808 | 0.900 | 0.961 | 0.986 | 0.998 | 1.000 |
| numeric_vars | L6 | 150 | 0.999/0.057 | 0.966 | 0.961 | 0.991 | 0.997 | 0.999 | 1.000 |
| misleading_iterator | L6 | 150 | 0.999/0.286 | 0.820 | 0.835 | 0.916 | 0.946 | 0.983 | 1.000 |

Table 20: Exploratory iterator activation-patching recovery for Qwen2.5-Coder-1.5B, cross-language transfer.

| condition | readout | pairs | clean/corrupt | L1 | L2 | L3 | L4 | L5 | L6 | L7 |
|---|---|---|---|---|---|---|---|---|---|---|
| Java | L7 | 10 | 0.989/0.709 | -0.300 | -0.146 | 0.616 | 0.867 | 0.845 | 0.940 | 1.000 |
| C++ | L7 | 11 | 0.979/0.769 | -0.066 | 0.025 | 0.785 | 0.869 | 0.869 | 0.904 | 1.000 |
| C | L7 | 0 | 0.000/0.000 | N/A | N/A | N/A | N/A | N/A | N/A | N/A |
| C# | L7 | 9 | 0.981/0.629 | 0.307 | 0.353 | 0.772 | 0.867 | 0.903 | 0.989 | 1.000 |
| JavaScript | L7 | 13 | 0.983/0.766 | 0.708 | 0.678 | 0.968 | 0.943 | 0.953 | 0.983 | 1.000 |
| PHP | L7 | 7 | 0.982/0.748 | -0.211 | -0.397 | 0.614 | 0.760 | 0.934 | 1.027 | 1.000 |

Table 21: Exploratory iterator activation-patching recovery for StarCoder2-7B, within-language perturbations.

| condition | readout | pairs | clean/corrupt | L1 | L2 | L3 | L4 | L5 |
|---|---|---|---|---|---|---|---|---|
| random_nouns | L5 | 150 | 0.998/0.569 | 0.931 | 0.766 | 0.966 | 0.951 | 1.000 |
| single_chars | L5 | 150 | 0.994/0.686 | 0.636 | 0.006 | 0.823 | 0.759 | 1.000 |
| all_same | L5 | 150 | 0.999/0.249 | 0.342 | 0.772 | 0.946 | 0.979 | 1.000 |
| numeric_vars | L5 | 150 | 0.998/0.530 | 0.889 | 0.811 | 0.953 | 0.955 | 1.000 |
| misleading_iterator | L5 | 150 | 0.997/0.628 | 0.777 | 0.104 | 0.921 | 0.843 | 1.000 |

Table 22: Exploratory iterator activation-patching recovery for StarCoder2-7B, cross-language transfer.

| condition | readout | pairs | clean/corrupt | L1 | L2 | L3 | L4 | L5 | L6 | L7 | L8 | L9 | L10 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Java | L10 | 150 | 0.996/0.750 | 0.254 | 0.634 | 0.829 | 0.913 | 0.957 | 0.964 | 0.979 | 0.988 | 0.994 | 1.000 |
| C++ | L10 | 150 | 0.996/0.771 | 0.337 | 0.756 | 0.892 | 0.938 | 0.966 | 0.969 | 0.979 | 0.988 | 0.995 | 1.000 |
| C | L10 | 0 | 0.000/0.000 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A |
| C# | L10 | 150 | 0.996/0.740 | 0.294 | 0.692 | 0.877 | 0.942 | 0.969 | 0.974 | 0.984 | 0.991 | 0.996 | 1.000 |
| JavaScript | L10 | 150 | 0.996/0.716 | 0.311 | 0.743 | 0.880 | 0.939 | 0.972 | 0.976 | 0.983 | 0.990 | 0.996 | 1.000 |
| PHP | L10 | 73 | 0.993/0.808 | 0.491 | 0.796 | 0.938 | 0.964 | 0.979 | 0.973 | 0.983 | 0.994 | 0.999 | 1.000 |

# C Exploratory boolean interventions

**Status and estimand.** These boolean interventions are exploratory diagnostics rather than evidence for the main causal claim. The committed export contains layer-wise means but not the per-case arrays needed to recompute clustered intervals, and the runs do not use the specified gate of the class/structure experiment. The exporter defines effect as $|\mathrm{clean} - \mathrm{intervened}|$ and specificity as that effect divided by the corresponding random-position-control effect. Specificity can become large when its denominator is small. We therefore report both specificity at the largest-effect layer and the maximum specificity over layers rather than selecting only the more favorable one.

Table 23: Exploratory boolean patch diagnostics. Effect and specificity use the exporter definitions above. No uncertainty interval is available from the committed summary.

| language | model | $n$ | peak layer | peak effect | spec. at peak | best spec. (layer) |
|---|---|---|---|---|---|---|
| C++ | Qwen2.5-1.5B | 157 | L0 | 14.845 | 3.6× | 7.9× (L20) |
| C++ | Qwen2.5-Coder-1.5B | 158 | L0 | 15.414 | 3.6× | 6.9× (L8) |
| C++ | StarCoder2-7B | 157 | L4 | 14.982 | 8.1× | 10.4× (L16) |
| Java | Qwen2.5-1.5B | 155 | L8 | 11.607 | 6.1× | 7.7× (L12) |
| Java | Qwen2.5-Coder-1.5B | 155 | L8 | 12.642 | 5.9× | 7.7× (L12) |
| Java | StarCoder2-7B | 155 | L4 | 13.900 | 7.5× | 8.6× (L12) |
| JavaScript | Qwen2.5-1.5B | 58 | L8 | 12.755 | 6.9× | 6.9× (L8) |
| JavaScript | Qwen2.5-Coder-1.5B | 62 | L8 | 12.717 | 6.6× | 7.4× (L12) |
| JavaScript | StarCoder2-7B | 129 | L4 | 12.137 | 8.1× | 8.2× (L16) |
| PHP | Qwen2.5-1.5B | 15 | L4 | 11.698 | 7.2× | 7.2× (L4) |
| PHP | Qwen2.5-Coder-1.5B | 16 | L4 | 11.425 | 4.7× | 7.4× (L0) |
| PHP | StarCoder2-7B | 17 | L4 | 11.545 | 6.3× | 9.3× (L16) |
| Python | Qwen2.5-1.5B | 176 | L8 | 12.948 | 6.0× | 7.2× (L16) |
| Python | Qwen2.5-Coder-1.5B | 176 | L8 | 13.275 | 6.7× | 7.8× (L16) |
| Python | StarCoder2-7B | 176 | L4 | 13.592 | 7.9× | 9.7× (L12) |

Table 24: Exploratory boolean steer diagnostics. Effect and specificity use the exporter definitions above. No uncertainty interval is available from the committed summary.

| language | model | $n$ | peak layer | peak effect | spec. at peak | best spec. (layer) |
|---|---|---|---|---|---|---|
| C++ | Qwen2.5-1.5B | 79 | L24 | 2.508 | 45.0× | 169.3× (L4) |
| C++ | Qwen2.5-Coder-1.5B | 79 | L24 | 1.436 | 602.0× | 602.0× (L24) |
| C++ | StarCoder2-7B | 78 | L20 | 0.653 | 6.9× | 345.4× (L4) |
| Java | Qwen2.5-1.5B | 78 | L24 | 2.486 | 72.2× | 826.9× (L8) |
| Java | Qwen2.5-Coder-1.5B | 78 | L24 | 1.712 | 72.0× | 179.9× (L0) |
| Java | StarCoder2-7B | 78 | L24 | 0.555 | 244.0× | 244.0× (L24) |
| JavaScript | Qwen2.5-1.5B | 31 | L24 | 1.763 | 24.5× | 76.8× (L8) |
| JavaScript | Qwen2.5-Coder-1.5B | 31 | L24 | 1.147 | 65.3× | 147.6× (L4) |
| JavaScript | StarCoder2-7B | 65 | L16 | 0.591 | 11.2× | 61.0× (L28) |
| PHP | Qwen2.5-1.5B | 7 | L24 | 2.759 | 190.2× | 190.2× (L24) |
| PHP | Qwen2.5-Coder-1.5B | 8 | L24 | 0.898 | 11.5× | 132.7× (L8) |
| PHP | StarCoder2-7B | 9 | L20 | 0.991 | 8.7× | 99.9× (L4) |
| Python | Qwen2.5-1.5B | 88 | L24 | 2.175 | 209.1× | 794.6× (L16) |
| Python | Qwen2.5-Coder-1.5B | 88 | L24 | 1.231 | 1027.1× | 1027.1× (L24) |
| Python | StarCoder2-7B | 88 | L20 | 0.721 | 8.6× | 55.1× (L4) |

**Omitted figures.** Per-cell diagnostic plots for the exploratory boolean interventions and the committed role analyses are reproducible from the same artifacts with `make_interp_appendix.py -full-figures`. They visualize the tables above and add no uncertainty estimate the exports lack.

Table 25: Exploratory boolean ablate diagnostics. Effect and specificity use the exporter definitions above. No uncertainty interval is available from the committed summary.

| language | model | $n$ | peak layer | peak effect | spec. at peak | best spec. (layer) |
|---|---|---|---|---|---|---|
| C++ | Qwen2.5-1.5B | 157 | L0 | 14.286 | 4.5× | 31.9× (L4) |
| C++ | Qwen2.5-Coder-1.5B | 158 | L0 | 12.927 | 4.4× | 76.2× (L20) |
| C++ | StarCoder2-7B | 157 | L0 | 13.405 | 4.3× | 36.5× (L4) |
| Java | Qwen2.5-1.5B | 155 | L16 | 9.720 | 28.5× | 86.0× (L12) |
| Java | Qwen2.5-Coder-1.5B | 155 | L4 | 9.331 | 12.3× | 62.4× (L12) |
| Java | StarCoder2-7B | 155 | L4 | 11.190 | 27.0× | 27.0× (L4) |
| JavaScript | Qwen2.5-1.5B | 58 | L16 | 10.852 | 26.7× | 26.7× (L16) |
| JavaScript | Qwen2.5-Coder-1.5B | 62 | L16 | 9.868 | 28.1× | 116.4× (L24) |
| JavaScript | StarCoder2-7B | 129 | L4 | 10.970 | 27.3× | 27.3× (L4) |
| PHP | Qwen2.5-1.5B | 15 | L16 | 15.698 | 15.0× | 102.4× (L20) |
| PHP | Qwen2.5-Coder-1.5B | 16 | L16 | 13.402 | 26.5× | 79.2× (L20) |
| PHP | StarCoder2-7B | 17 | L4 | 14.260 | 19.4× | 19.7× (L8) |
| Python | Qwen2.5-1.5B | 176 | L0 | 12.144 | 5.8× | 36.7× (L24) |
| Python | Qwen2.5-Coder-1.5B | 176 | L0 | 11.286 | 6.8× | 33.1× (L16) |
| Python | StarCoder2-7B | 176 | L4 | 11.818 | 26.5× | 26.5× (L4) |

# D Explicit coverage holes

- Accumulator and index/key results use legacy exports. No committed problem-grouped, five-seed CSV reproduces those roles under the modern protocol.
- Iterator has a model-free surface comparator. The probe-minus-surface gap is not a paired clustered interval.
- Class/structure transfer exports include Python, C++, JavaScript, and C, but not Java, C#, or PHP.
- Boolean probe exports include Python, Java, JavaScript, and PHP. C, C#, and C++ are absent, and renaming is available only for Python.
- Gate-specified class/structure patching stopped after the Qwen2.5-1.5B null. Coder, StarCoder2, and the all-layer core sweep were not attempted.
- No matched model-free surface comparator is committed for the class/structure role.

# E Reproducibility

The appendix is generated from committed aggregate boolean, iterator, class or structure, and patching artifacts with `uv run python scripts/make_interp_appendix.py`. The claim audit uses `uv run python tests/test_interp4d_claims.py`. Modern probes use problem hashes, validation-selected layers, and five seeds where stated. The class or structure intervention records model and dataset revisions, hashes, all $4{,}032$ rows across its primary and behavior schedules, and clustered intervals. The paired boolean CSV is aggregate evidence because its aligned prediction inputs were not retained. The regression test checks manuscript-to-artifact consistency rather than reproducing activation extraction or probe fitting.

# F Broader impacts

Positive effects are scientific. Probe scores near $0.98$ are easy to read as evidence that a model computed a role. If such claims later inform audits of coding assistants, an outcome-aware record reduces that overclaim.

Negative effects follow from the same scores. A reporting matrix can be treated as permission to stop at decodability if the bound is ignored. Role labels derived from public competitive-programming solutions could be reused to train name-based heuristics that recover the label from the identifier. No new generative model is released. The source programs were already public.

## G Existing assets and licenses

Source programs come from XLCoST under Apache 2.0 [Zhu et al., 2022]. Qwen2.5-1.5B and Qwen2.5-Coder-1.5B are used under Apache 2.0. StarCoder2-7B is used under the BigCode OpenRAIL-M license. Each release is cited by name and identifier. Derived role labels inherit Apache 2.0 from XLCoST.

## H New assets

The new assets are structurally labeled XLCoST variable-role data and the analysis scripts that produce the reported tables. A dataset card documents fields, extractors, splits, and the parent license. Committed result files carry artifact provenance. An anonymized snapshot is at `https://anonymous.4open.science/r/SameScoreDifferentEvidence2026`.

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